

Gene function in mouse embryogenesis: get set for gastrulation

Patrick P. L. Tam and David A. F. Loebe

Abstract | During early mouse embryogenesis, temporal and spatial regulation of gene expression and cell signalling influences lineage specification, embryonic polarity, the patterning of tissue progenitors and the morphogenetic movement of cells and tissues. Uniquely in mammals, the extraembryonic tissues are the source of signals for lineage specification and tissue patterning. Here we discuss recent discoveries about the lead up to gastrulation, including early manifestations of asymmetry, coordination of cell and tissue movements and the interactions of transcription factors and signalling activity for lineage allocation and germ-layer specification.

Conceptus

All structures that develop from fertilization, including embryonic and extraembryonic structures.

Implantation

The attachment of the embryo to the uterine wall. In the mouse, this occurs about 4 days post coitum as the first step in establishing the maternal–fetal connection.

Blastocyst

The structure that the embryo adopts after the morula stage, 3.5–4.5 days post coitum, consisting of the trophoblast surrounding an inner cell mass and blastocoel cavity.

Embryology Unit, Children's Medical Research Institute and Faculty of Medicine, University of Sydney, Westmead, NSW 2145, Australia.

Correspondence to P.P.L.T. e-mail:

ptam@cmri.usyd.edu.au
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A dramatic transformation of the architecture of the mouse conceptus takes place immediately after implantation. The implanted blastocyst is a vesicular structure comprising an inner cell mass (ICM) inside the trophoblast (FIG. 1a). Within 3 days, the polar trophoblast and the ICM develop into an elongated structure that is made up of the ectoplacental cone (which connects the embryo to the uterus), the extraembryonic ectoderm (ExE), the epiblast and a layer of visceral endoderm (FIG. 1b–d). Gastrulation then commences with the formation of the primitive streak (FIG. 1e), through which epiblast cells ingress to form the mesoderm and the endoderm. These two tissues and the ectoderm (the descendants of epiblast cells that do not pass through the primitive streak) constitute the three primary germ layers (BOX 1) that contain progenitors of all tissues^{1,2}. Recent studies on the early post-implantation embryo have revealed complex patterns of molecular activity in the derivatives of the ICM and the trophoblast; this indicates that embryonic polarity and tissue patterning emerge well before gastrulation. These studies have also shown that the extraembryonic tissues, which are unique to mammalian embryos, are not just a connection between maternal and embryonic tissue — they are crucial for embryonic patterning.

This Review focuses on the spatial and temporal deployment of transcriptional regulation and intercellular signalling (Supplementary information S1 (table)) during development from the blastocyst to the gastrula. The information we discuss comes from the developmental defects of mutant mouse embryos and direct experimentation on embryos that were grown outside the uterine environment, involving cell imaging, microsurgical manipulation and alterations to gene

activity using molecular biological techniques (BOX 2). Here we discuss the complexity of gene functions for constructing a mouse gastrula. Focusing on important recent findings, we will first discuss the origin of the cell lineages in the implanting embryo and the molecular basis of embryonic and extraembryonic tissue patterning, with particular emphasis on the role of tissue interactions. We will then highlight the signalling and transcription-factor activities that are required for the regulation of cell movement and the formation of the primary germ layers.

Generation of cell lineages at implantation

Maintaining the pluripotency of the epiblast. Before lineage allocation occurs, the precursor populations in the epiblast must be maintained in an undifferentiated, pluripotent state in order to generate the full range of cell types that make up the body. Maintenance of pluripotency requires interactions between transcription-factor complexes. Two such complexes are involved, one including **POU5F1** (POU domain class 5 transcription factor 1, also known as OCT4) and **SOX2** (SRY box 2), and a second involving **nanog** and **SALL4** (sal-like 4). Pluripotency is maintained by the activity of the two complexes on downstream genes, simultaneously promoting expression of the genes that encode their own and each other's constituent members^{3–8}. Polycomb proteins provide a further layer of regulation by suppressing the expression of genes that promote differentiation^{9,10}. Maintaining pluripotency of the epiblast also requires signalling by **nodal**, a transforming growth factor- β (TGFB)-related factor. In embryos that lack nodal, the epiblast differentiates prematurely to neuroectoderm^{11,12}. Nodal signalling through the intracellular effectors

Extraembryonic

Structures in the conceptus that do not contribute to the fetal tissues, including those derived from the trophoblast and primitive endoderm. These tissues support the growth of the embryo and are a source of signals for patterning.

Epiblast

A structure in the post-implantation embryo that develops from the inner cell mass, from which the ectoderm, mesoderm and definitive endoderm are derived.

Gastrulation

A stage of development in which the primary germ layers are formed (ectoderm, mesoderm and definitive endoderm), from which all the fetal tissues will develop.

Primitive streak

A structure that forms in the posterior region of the embryo and is the first visible sign of gastrulation. Epiblast cells ingress through the primitive streak and are allocated to the mesoderm or definitive endoderm.

Ingress

The process of cells entering the primitive streak.

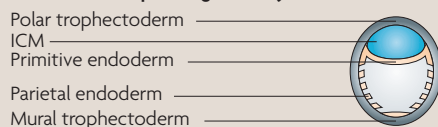
Primary germ layers

The ectoderm, mesoderm and definitive endoderm cell layers that form at gastrulation, from which all fetal and some extraembryonic tissues develop.

Potency

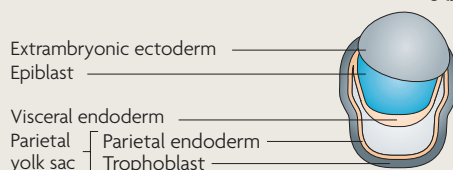
The ability of a cell to differentiate into one or more cell types.

a E4.0–4.25: Implanting blastocyst



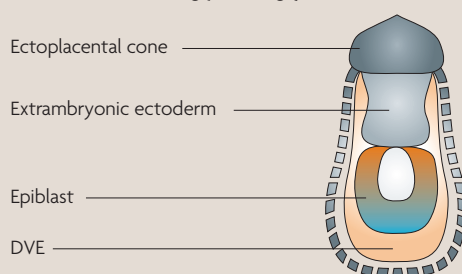
- Proliferation of the polar trophoctoderm and the ICM
- Sorting the precursors of the primitive endoderm from the ICM
- Organization of the primitive endoderm into an epithelium on the blastocoelic surface of the ICM
- Primitive endoderm then contributing cells to form the parietal endoderm

b E5.0: No localized visceral-endoderm thickening (pre-DVE stage)



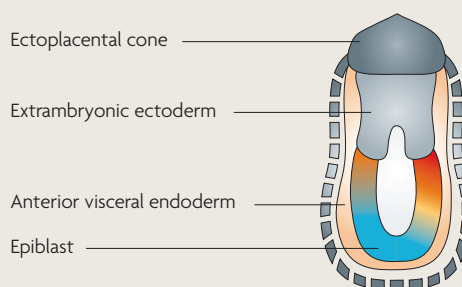
- Proliferation of the extraembryonic ectoderm and formation of the ectoplacental cone
- Primitive endoderm partitioned into visceral and parietal endoderm
- Proliferation of the epiblast, which is derived from the ICM

c E5.5: DVE thickening (DVE stage)



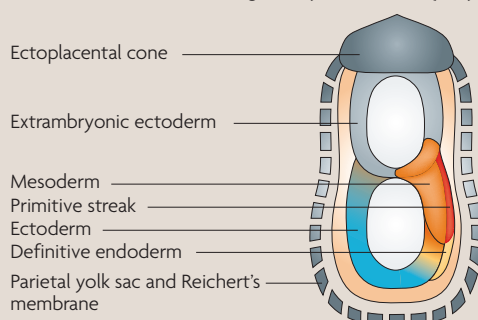
- Cavitation and epithelial morphogenesis of the epiblast
- Proximal–distal patterning of the epiblast
- Specification of the precursor of anterior visceral endoderm in the distal region of the visceral endoderm

d E6.0: Anterior visceral-endoderm or lateral visceral-endoderm thickening (post-DVE stage)



- Directional movement of the DVE cells to form the anterior visceral endoderm
- Positioning of the primitive streak
- Anterior–posterior patterning of the epiblast

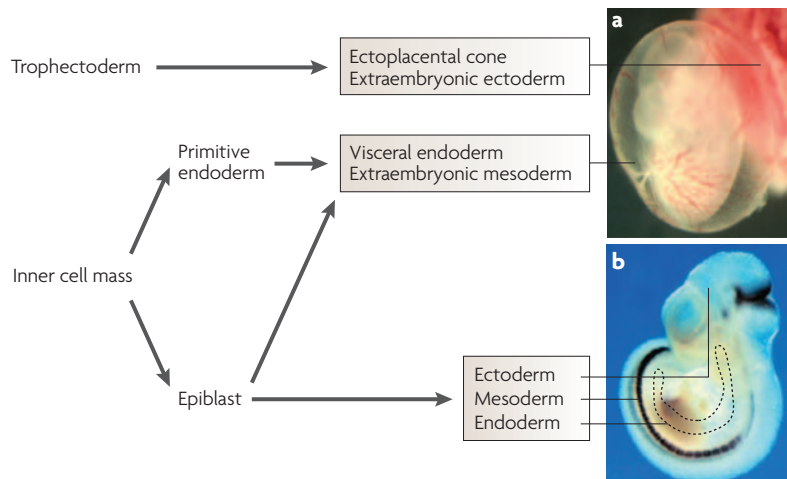
e E6.5: Primitive-streak and germ-layer formation (early primitive-streak stage)



- Gastrulation:
- Formation of the primitive streak
 - Allocation of germ-layer precursors
 - Morphogenetic cell and tissue movement

Figure 1 | Early post-implantation development in the mouse. Tissue types in the embryo are shown at five developmental stages. **a** | Implanting blastocyst. **b** | Pre-DVE (distal visceral endoderm). **c** | DVE. **d** | Post-DVE and pre-primitive streak. **e** | Early primitive streak. The lineage relationship of the derivatives of the trophoctoderm and inner cell mass (ICM) is described in BOX 1. The primitive endoderm is coloured grey. The ICM in panel **a** and the epiblast in panel **b** are coloured blue. The multi-colouring of the epiblast in panels **c–e** indicates the regionalization of prospective germ-layer progenitors: ectoderm (coloured blue), mesoderm (coloured orange), definitive endoderm (coloured yellow) and the primitive streak (coloured crimson) are shown according to the fate maps². In panel **c**, orange represents the mesendoderm. In panel **a**, cells of the primitive endoderm colonize the blastocoelic surface of the mural trophoctoderm and become the parietal endoderm, which, together with the trophoblast that derives from the mural trophoctoderm, forms the parietal yolk sac in panel **b**. As the epiblast and extraembryonic ectoderm become elongated into a cylinder, the visceral endoderm covers their outer surface. The trophoblast and parietal endoderm, together with the thick basement membrane between them (the Reichert's membrane) surround the entire conceptus. These are removed (indicated by a dotted line) in panels **c–e**. Nomenclature for staging is adapted from REFS 12,63,122.

Box 1 | Allocation of extraembryonic and germ-layer derivatives



Key cell lineages are delineated shortly before implantation. At the morula stage, when the embryo consists of a ball of about 16 cells, the outer cells express caudal type homeobox 2 (CDX2) and develop epithelial characteristics¹¹⁴. These cells become the trophoblast, from which the extraembryonic ectoderm and ectoplacental cone that make up part of the placenta are derived, shown in **a**. Cells on the inside express genes that are characteristic of pluripotent cells and form the inner cell mass (ICM) of the blastocyst, from which the epiblast derives. The ICM gives rise to the primitive endoderm, which produces the visceral endoderm lining the extraembryonic yolk sac.

The epiblast consists of a population of pluripotent cells from which all the fetal tissues will form, as well as the extraembryonic mesoderm of the yolk sac, the allantois and the amnion. Before gastrulation, the body plan of the embryo is mapped onto the epiblast in gene-expression domains that correspond to the prospective anterior–posterior embryonic axis. At gastrulation, this blueprint is translated into three dimensions by the allocation of cells to the three germ layers — the ectoderm, mesoderm and endoderm.

Cells from the epiblast are recruited into the primitive streak and migrate out, after which they either form a new layer of mesoderm or are incorporated into the endoderm to form the definitive (gut) endoderm (outlined in **b**). Cells that remain in the epiblast form the ectoderm, which gives rise to the surface ectoderm and the neural tissues, also indicated in **b**. Clonal analysis shows that epiblast cells are not restricted in their lineage potency prior to recruitment and ingression^{115,116}. It is likely that a transient cell population, known as the mesendoderm, which has the potential to form mesoderm or definitive endoderm, exists within the primitive streak. Mesendodermal cells exercise a lineage choice between endoderm and mesoderm in the primitive streak. Although these cells have not been directly identified in the mouse embryo, evidence for a bipotential cell population comes from experiments with embryonic stem (ES) cells. Mouse ES cells can be induced by activin to differentiate into mesendodermal progenitors, with the dual potential to form mesoderm and endoderm^{93,117}.

Cells that are allocated to the mesoderm are distributed to four main tissue compartments: the axial mesoderm (notochord); the paraxial mesoderm, which forms the somites (the darkly stained tissue in **b**, which shows a whole mount *in situ* hybridization for *Engrailed 1* (*En1*) expression) and subsequently the vertebral column and skeletal muscle; the lateral plate mesoderm (circulatory system and gut wall); and the extraembryonic mesoderm (yolk sac vasculature and blood). Fate mapping of the primitive streak shows that cells of the axial mesoderm and the anterior definitive endoderm of the embryonic gut ingress through the anterior-most region, the paraxial mesoderm emerges in the anterior-third region, the lateral plate mesoderm through the middle region, and those of the extraembryonic mesoderm through the posterior region¹¹⁸. During gastrulation, mesoderm cells that are recruited earlier are allocated to more anterior sites in their respective tissue compartment than those that are recruited later. In essence, the site and timing of recruitment and ingression of cells directly influence the allocation to specific tissue types and the destination of these cells in the anterior–posterior axis^{119,120}.

The definitive endoderm forms the epithelium of the gut tube, lungs and organs including the liver, pancreas and thyroid. The precursors of the definitive endoderm ingress through the anterior part of the primitive streak, and the definitive endoderm seems to be regionalized early in its development. This is reflected by the lack of ‘pan-endoderm’ gene markers, which may be expressed throughout the definitive endoderm⁸⁶.

SMAD2 and **SMAD3** is also required for maintaining pluripotency in human embryonic stem (ES) cells¹³.

There is conflicting evidence regarding the role of WNT signalling in regulating cell-lineage differentiation. Inhibition of **GSK3** (glycogen synthase kinase 3), which enhances WNT signalling by increasing the nuclear accumulation of β -catenin, promotes maintenance of the undifferentiated state of ES cells¹⁴. However, pharmacological GSK3 inhibitors are likely to have other effects, raising the possibility that pluripotency might be maintained by disrupting other unknown pathways. Conversely, in mouse embryos the epiblast will differentiate precociously into mesoderm-like cells when WNT-downstream genes are upregulated by constitutively active β -catenin¹⁵, and in the absence of **Wnt3** (Wingless-related MMTV integration site 3)¹⁶ the epiblast continues to express the pluripotency marker *Pou5f1*. In agreement with this, inhibition of endogenous WNT signals in mouse ES cells by **DKK1** (dickkopf 1) prevents the cells from differentiating into mesoderm¹⁷. On balance, the data indicate that maintenance of pluripotency requires suppression of WNT signalling, which promotes differentiation, but the possibility that WNT signalling has an additional role in maintaining pluripotency remains.

Allocation and formation of the primitive endoderm. By the time the embryo implants into the uterus, a third cell lineage, the primitive endoderm (PrE), has formed on the blastocoelic surface of the ICM, from which it derives (FIG. 1a). The PrE then forms both the parietal endoderm, which lines the blastocoelic surface of the mural trophoblast (FIG. 1a,b), and the visceral endoderm, which forms an epithelium on the surface of the epiblast and ExE (FIG. 1c–e). Several key questions regarding the origins and development of the PrE and its derivatives remain to be answered. When do the PrE precursors first appear? Where in the ICM do the PrE cells arise? How do they become an epithelium? What is the source of patterning information for regionalizing the PrE and subsequently the visceral endoderm, and how early is regionalization detectable? Recently, some progress has been made in finding answers to these questions.

It is not known if the allocation of ICM cells to the PrE is influenced by their position in the cell cluster. A recent gene-expression study has shown intermingling populations of *Nanog* and **Gata6** (GATA-binding protein 6)-expressing cells in the ICM¹⁸. By tracing the descendants of individual ICM cells, it was shown that they can contribute to either the epiblast or the PrE, but not both¹⁸. Although it has yet to be formally proven, the finding implies that the *Gata6*-positive cells are the progenitors of the PrE, and the *Nanog*-expressing cells presumably form the epiblast. **GRB2** (growth factor receptor bound protein 2, a receptor-binding adaptor protein and member of mitogen-activated protein (MAP) kinase kinase kinase signalling pathways) is required for expression of *Gata6* in the ICM and the formation of the PrE^{18,19}, indicating a role for receptor tyrosine kinase signalling in the early stages of PrE formation.

GATA6 and its transcriptional targets, *Hnf4a* (hepatocyte nuclear factor- α , a regulator of tight junction formation and epithelial cell polarity^{20,21}) and *Dab2* (disabled homologue 2, a low-density lipoprotein receptor adaptor protein^{22–24}) are also involved in the formation of the visceral endoderm. Although PrE formation is initiated in *Gata6*^{−/−} and *Hnf4a*^{−/−} blastocysts, visceral-endoderm formation is disrupted^{21,25–27}. The visceral endoderm of

Dab2^{−/−} embryos is disorganized, and cells that express markers of the PrE and visceral endoderm are internalized in the epiblast²⁸. *Dab2* function is therefore crucial for morphogenesis, but not for specification of the visceral endoderm. Laminin B1 subunit might also have a role in the formation of the visceral endoderm — it is involved in cell adhesion and is upregulated in ES cells that express *Gata6* (REF. 29). Together, these studies indicate that the sorting of PrE cells from other ICM cells begins with receptor tyrosine kinase signalling-mediated regulation of *Gata6* expression, which in turn activates other genes that are involved in cell sorting and formation of an epithelial architecture. However, the precise mechanism by which cells of the ICM choose between the PrE lineage and other cell types is unclear.

Signalling activity maintains the ExE, which is formed by the proliferation of the polar trophoctoderm and is closely apposed to the epiblast (FIG. 1b,c). Therefore, there is potential for reciprocal interactions in the control of the patterning and differentiation of both tissues. The proliferation and differentiation of the ExE requires fibroblast growth factor (FGF) signals from the ICM³⁰. *Fgf4*^{−/−} embryos cannot maintain normal trophoctoderm development beyond the blastocyst stage³¹. The response by the ExE to FGF4 is likely to be mediated, at least in part, by FRS2 (FGF-receptor substrate 2)³², a membrane-bound docking protein. Consistent with defective ExE differentiation, trophoctodermal stem cells could not be derived from *Frs2*^{−/−} embryos *in vitro*³². A possible link between FGF and nodal signalling is indicated by the upregulation of *Fgf4* in response to nodal signalling³⁰ (FIG. 2a). As *Frs2* activity seems to be required to maintain normal *Nodal* expression levels³², nodal and FGF4 from the epiblast might therefore work together to maintain a progenitor population in the ExE, preventing these cells from undergoing premature differentiation to trophoblasts³⁰.

Regionalization of the PrE and visceral endoderm

The visceral endoderm, which surrounds the epiblast and ExE (FIG. 1c–d), is an important source of signals for embryonic patterning. In particular, the anterior visceral endoderm (AVE) (FIG. 1d) is crucial for correct anterior–posterior patterning³³, functioning as a source of antagonists for posteriorizing signals such as nodal and WNT^{34,35}. The precursors of the AVE arise at the distal tip of the embryo, and are termed the distal visceral endoderm (DVE). These cells then migrate to the anterior side of the embryo to form the AVE (FIG. 1c). On the opposite side, interactions between the posterior visceral endoderm and the epiblast near the border between the embryonic visceral endoderm (EmVE) and the extraembryonic visceral endoderm (ExVE) can influence gene expression in the primitive streak³⁶. The formation of these localized signalling centres requires the establishment of regionalized gene-expression domains within the visceral endoderm.

Is there asymmetry in the PrE? Asymmetry of gene expression occurs before the appearance of morphologically distinguishable anterior–posterior patterning. However, when this is first detectable is the subject of

Box 2 | Tools for analysis of gene function in mouse embryos

Loss- or gain-of-function mutants

- Spontaneous mutations that cause a change of gene function; for example, the *Apc*^{min} mutant, which exhibits constitutive β -catenin activity³⁸, and the brachyury mutant, which shows loss of transcription factor function.
- Chemically induced mutations, such as those that are generated in screens using N-ethyl-N-nitrosourea (ENU) to induce point mutations; for example, the *Nckap1* (NCK-associated protein 1) mutant, which disrupts cell migration⁶⁶.
- Gene-trap mutagenesis, in which a reporter construct inserts at random locations throughout the genome; for example, the WNT co-receptor gene *Lrp6* (low-density lipoprotein receptor-related protein 6).
- Targeted mutations that result in either complete removal of all or part of the coding region, or conditional knockouts, in which a region of the gene is flanked by *loxP* or *frt* recombinase target sites. Excision is catalysed by crossing with a strain that harbours CRE or FLP recombinase with tissue- or stage-specific expression; for example: epiblast-specific SMAD2 and SMAD3 mutants^{91,92}, primitive-streak and endoderm-specific β -catenin mutant¹²¹.

Dominant negative or modified sequence

A modified form of a protein is expressed, which either disrupts the function of a wild-type protein or is modified in some way to alter its function. This includes constitutively active forms or precursor proteins that are resistant to cleavage; for example, replacement of the wild-type *Nodal* gene with a cDNA that encodes a cleavage-resistant nodal pro-protein⁵⁵.

Knock-in: replacement or insertion into a gene of interest

This includes rescue of a mutant phenotype by substitution of a sequence of the ORF of another gene, or by introduction of a reporter such as β -galactosidase or GFP into the gene; for example, replacement of *Otx2* (orthodenticle homologue 2) coding sequences with a *Dkk1* (Dickkopf 1) cDNA⁶⁷.

Transgenic reporter

A construct that contains a reporter under the control of tissue-specific promoter or enhancer elements is introduced into the genome. Expression of the transgene should reflect that of the endogenous gene that is controlled by these regulatory elements; for example, demonstration of asymmetry in the primitive endoderm by left–right determination factor 1 (*Lefty1*)–*lacZ* expression, and visualization of gene expression by *Nodal*–*lacZ* and MIX1–homeobox-like 1 (*Mixl1*)–*GFP* reporters.

RNAi

This might involve the introduction of small interfering RNA (siRNA) molecules into early embryos by electroporation, or stable expression of a short hairpin RNA (shRNA) construct in embryonic stem (ES) cells, which can then be used to create embryos for phenotypic analysis; for example, electroporation of BMP4 (bone morphogenetic protein 4) siRNA into early embryos⁴³, and shRNA knockdown in ES-cell-derived embryos in tetraploid complementation assays⁷³.

Creation of completely ES-cell-derived embryos

This method takes advantage of the inability of tetraploid cells to contribute efficiently to embryonic tissues. ES cells are introduced into a tetraploid host embryo by aggregation or injection and transferred to a surrogate mother. The tetraploid cells contribute to the extraembryonic lineages, but are excluded from the embryo itself. This contrasts with the conventional approach of making chimaeras by introducing ES cells into a diploid host embryo, which results in co-contribution of host and ES-derived cells to the embryonic tissues; for example, the demonstration that embryos that are derived from *Hnf4*^{−/−} (hepatocyte nuclear factor) ES cells progress through gastrulation if they are complemented by tetraploid wild-type visceral endoderm²⁶.

a Visceral endoderm

b Epiblast

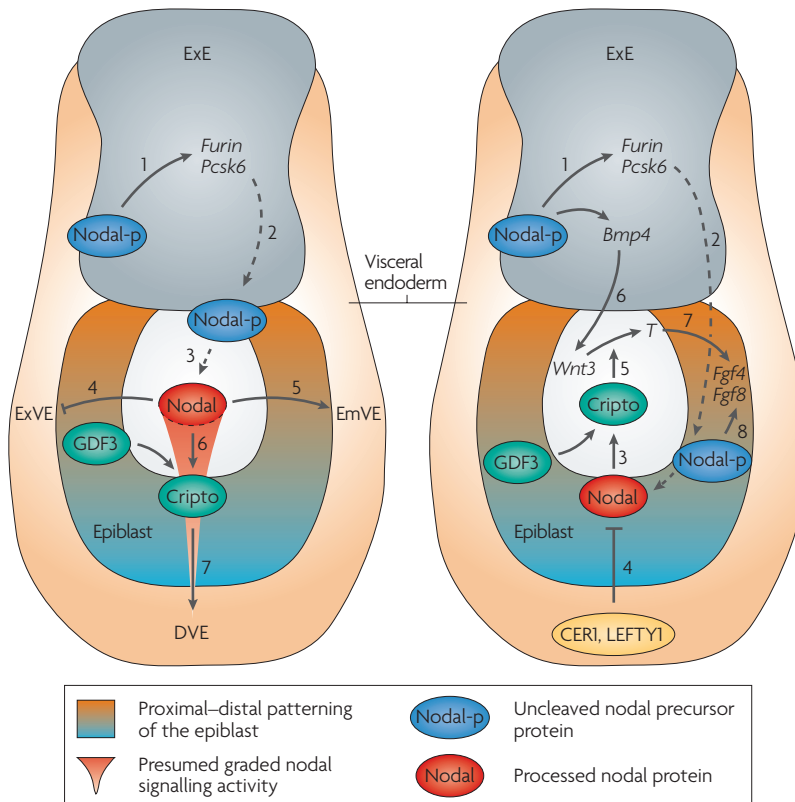


Figure 2 | A network of signalling activity leading to patterning of the visceral endoderm and the epiblast in the distal visceral endoderm (DVE)-stage embryo. The relational network is based on the data obtained by studying the effects of loss of function of nodal or other signalling cascade components and cleavage-resistant nodal precursor protein. Signalling function involves the production, translocation and processing of the signal protein, followed by positive and negative transcriptional regulation of gene expression. Gene products are circled and gene transcripts are in italics. The various steps of the network are numbered serially. **a** | Patterning of the embryonic visceral endoderm (EmVE). In the extraembryonic ectoderm, nodal precursor protein, which is produced by the epiblast, acts by activin receptors 1B and 2A to activate the transcription of *Furin* and *Pcsk6* (step 1), which encode convertase enzymes. The extraembryonic ectoderm (ExE) then secretes the convertase enzymes encoded by *Furin* and *Pcsk6* (step 2), which cleave the nodal precursor (step 3) to produce the processed form of nodal. Nodal signals from the epiblast repress genes such as *Hnf4* (hepatocyte nuclear factor), *Gata4* (GATA-binding protein 4), *Ttr* (transferrin) and *Furin* (step 4), which are normally expressed in the extraembryonic visceral endoderm (ExVE) after the distal visceral endoderm (DVE) stage. Simultaneously (step 5), nodal signals maintain the expression of embryonic visceral endoderm (EmVE)-related genes such as *Lhx1* (LIM-domain homeobox 1), *Fgf5* (fibroblast growth factor 5), *Fgf8*, *Bmp2* (bone morphogenetic protein 2), *Otx2* (orthodenticle homologue 2) and *Foxa2* (forkhead box A2). Nodal and GDF3 (growth differentiation factor 3) working with cripto, a co-receptor, generate a proximal–distal gradient of signalling activity in the embryo (step 6), which is crucial for the specification of the DVE (step 7). Signals from the ExE, including BMP4 (not shown), are also required for regionalization of the visceral endoderm and proper formation of the DVE. **b** | Patterning of the epiblast. In the ExE, the nodal precursor protein activates the expression of *Bmp4* (step 1), in addition to the convertase enzyme genes *Furin* and *Pcsk6*. Similar to the signalling process in patterning the visceral endoderm, the convertases process the nodal protein (step 2). A graded signalling activity, which is mediated by nodal, GDF3 and cripto (step 3), and antagonized by LEFTY1 (left–right determination factor 1) and CER1 (cerberus 1 homologue) from the DVE (step 4), upregulates the expression of mesoendoderm-related genes in the proximal epiblast (step 5). BMP4 signal from the ExE activates a regulatory pathway involving *Wnt3* (step 6), which activates *T* (brachyury) expression, initially near the epiblast–ExE region and later in the region of the forming primitive streak. Expression of FGFs is downstream of *T* (step 7), and is enhanced by the activity of the nodal precursor protein (step 8).

some controversy. Asymmetrical gene expression has been observed in the visceral endoderm of embryos at the DVE stage (FIG. 1c). The nodal antagonists *Cer1* (cerberus 1 homologue) and *Lefty1* (left–right determination factor 1) are more highly expressed on one side (presumably the anterior) of the DVE³⁷. Similarly, next to the border of the epiblast and ExE, *Wnt3* is expressed asymmetrically³⁶.

However, asymmetry might be established in the PrE shortly after implantation, before formation of the visceral endoderm. One recent study showed transiently asymmetrical expression of *Hhex* (haematopoietically expressed homeobox) and *Rhox5* (reproductive homeobox 5, also known as *Pem*)³⁸ but no asymmetrical expression of *Lefty1*. By contrast, two other studies indicate consistent and asymmetrical *Lefty1* expression in the PrE. Expression of a *Lefty1-lacZ* transgene³⁹ and the endogenous *Lefty1* gene was observed to be skewed towards one side of the PrE^{12,39}, and seen to continue in the visceral endoderm¹². Although this finding has been disputed³⁸, the discrepancy might be partly explained by the choice of probes. If the probe that was used in the study by Chazaud and colleagues³⁸ detects not only *Lefty1*, it is possible that this non-specific detection could have obscured the true *Lefty1* expression domain. Intriguingly, asymmetrical *Lefty1-lacZ* expression was detected in blastocysts that were grown *in vitro* without contact with uterine tissue or any physical substrate. If the asymmetrical expression of *Lefty1* is ultimately confirmed, this raises the possibility that embryonic asymmetry might be acquired independently of any interaction with the uterus that accompanies implantation³⁹.

Proximal–distal patterning of the visceral endoderm. Before the appearance of anterior–posterior patterning, the visceral endoderm becomes polarized in the proximal–distal dimension. Between the pre-DVE stage and the DVE stage (FIG. 1b,c), differences in gene expression emerge between the EmVE, which surrounds the epiblast, and the ExVE, which surrounds the ExE. Recently, a role for nodal signalling in demarcating these two regions has been demonstrated (FIG. 2a, steps 4,5). In *Nodal*^{-/-} embryos, genes that are normally downregulated in the EmVE at embryonic days E5.5 and E6.5, continue to be expressed beyond this stage¹². This indicates that an early function of nodal signalling is to divide the visceral endoderm into embryonic and extraembryonic compartments. The visceral endoderm is further subdivided when a population at the distal tip of the embryo becomes specified as the DVE¹² (FIG. 1c; FIG. 2a, steps 6,7; TABLE 1). Subsequently, the DVE is relocated to the prospective anterior region of the conceptus to form the AVE (FIGS 1d,3a).

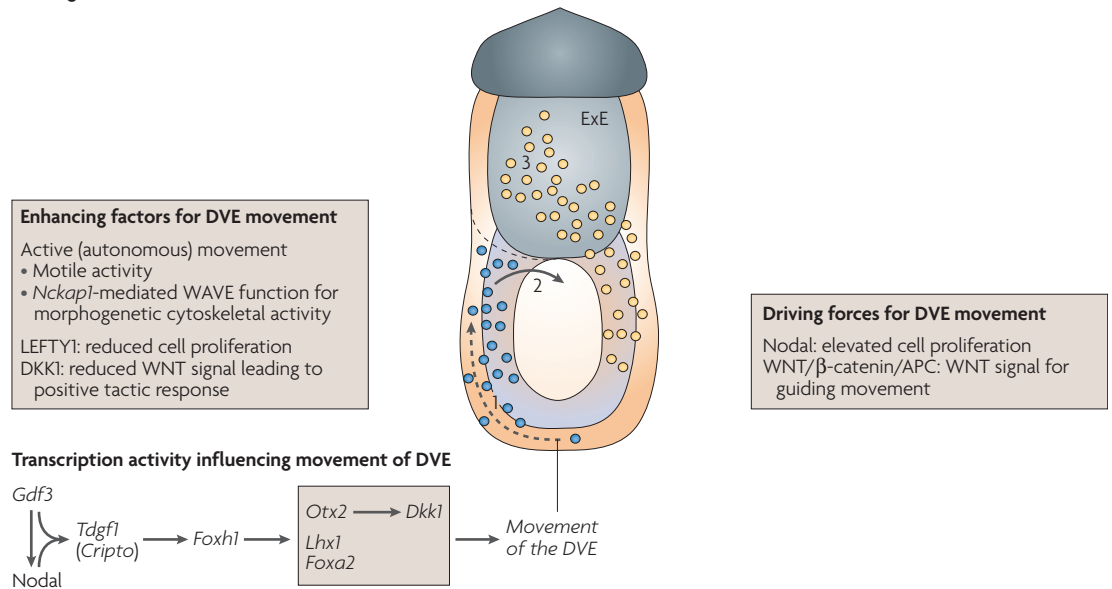
Interactions with the ExE and the epiblast underpin the formation of the DVE. The ExE is a source of signals for patterning the visceral endoderm. Separation of the ExE from the embryonic (epiblast and EmVE) fragment results in an expansion of the expression domains of DVE markers^{40,41}. However, in *Elf5*^{-/-} (E74-like factor 5) embryos, which lack an ExE and therefore have the

Table 1 | Mutations that affect patterning of the visceral endoderm and the epiblast

Gene	Genetic modification	DVE formation	DVE migration	Epiblast patterning	Germ-layer potential	Refs
Nodal signalling						
<i>Nodal</i>	KO	No	-	No primitive streak	Ectopic and expanded neuroectoderm	11,12, 47
	Conditional KO — mosaic in epiblast derivatives	Yes	Variable	Forms primitive streak; anterior (head) truncation	Forms anterior mesoderm but deficient in definitive endoderm	123
	Pro-viral insertion	-	-	Lacks any tissue patterning	Some mesoderm forms	124
	Hypomorphic	Yes	Defective	Defective	Normal	125
	Protease-cleavage resistant	-	-	Enlarged primitive streak	Normal	55
<i>Smad2</i>	KO and epiblast-conditional KO	No	-	Posteriorized	Forms extraembryonic mesoderm	47
<i>Cripto</i>	KO	Yes	No	No primitive streak	No mesoderm, ectopic or expanded neuroectoderm	50
<i>Foxh1</i>	KO	Yes	Yes, for the majority; No, in the most severe mutant	Appropriate anterior–posterior patterning in the less severe mutant; persistent proximal–distal patterning in the most severe mutant	Lacks axial mesoderm, forms paraxial and extraembryonic mesoderm	51,88
<i>Cer1, Lefty1</i>	Double KO	-	-	Expanded anterior primitive streak or multiple primitive streaks	Expanded domain of anterior primitive streak derivatives	56
<i>Drap1</i>	KO	-	-	Expanded primitive streak	Excess mesoderm	57
<i>Otx2</i>	KO	Yes	No, but can be rescued by DKK1 expression or reduced β -catenin	Anterior truncation	Loss of anterior neuroectoderm derivatives	35,67
BMP/TGFB signalling						
<i>Bmp4</i>	KO	-	-	Variable — no primitive streak in some embryos	No mesoderm forms	52
	KD	Defective	Defective	Primitive streak forms	Reduced mesoderm	43
<i>Bmpr1</i>	KO	-	-	No primitive streak	No mesoderm	53
<i>Bmpr2</i>	KO	-	-	No anterior–posterior patterning	No mesoderm	54
<i>Gdf3</i>	KO	Defective	No	Defective primitive-streak formation	Reduced mesoderm and definitive endoderm	98
WNT signalling						
<i>Wnt3</i>	KO	Yes	Yes	No primitive streak, no anterior neural patterning	No mesoderm forms	16
<i>Ctnnb1</i>	KO; Primitive-streak KO; Chimaera: epiblast	Yes, <i>Cer1</i> -positive cells present but no <i>Hhex</i> - or <i>Hesx1</i> -expressing cells	No, for <i>Cer1</i> -expressing cells	No primitive streak; defective anterior–posterior patterning	No mesoderm forms	45,49, 121
<i>Lrp5, Lrp6</i>	Double KO	Yes	Yes	No primitive streak	No mesoderm or definitive endoderm forms	59
<i>Macf1</i>	KO	Yes	-	No primitive streak	Neural plate forms, no mesoderm	126
<i>Mesd</i>	Radiation induced	Yes	Yes	Proper proximal–distal pattern	No primitive streak, mesoderm or organizer, expanded <i>Otx2</i> , <i>Cripto</i> and <i>Pou5f1</i> domain in the epiblast	127
<i>Apc</i>	<i>min</i> mutation; gain of WNT function	No	-	Acquires proximal–posterior characteristics	Mesoderm forms	38
Cytoskeletal regulation						
<i>Nckap1</i>	KO	Yes	No (some embryos)	Axis duplication	Normal	66

Dash (-) indicates not directly tested. *Apc*, adenomatous polyposis coli; BMP, bone morphogenetic protein; *Cer1*, cerberus 1 homologue; *Ctnnb1*, β -catenin gene; DKK1, dickkopf 1; *Drap1*, DR1-associated protein 1; DVE, distal visceral endoderm; *FoxH1*, forkhead box H1; *Gdf3*, growth differentiation factor 3; *Hesx1*, homeobox gene expressed in ES cells; *Hhex*, haematopoietically expressed homeobox; KO, knockout; KD, siRNA knockdown; *Lefty1*, left–right determination factor 1; LRP, low-density lipoprotein receptor-related protein; *Macf1*, microtubule actin cross-linking factor 1; *Mesd*, mesoderm development; *Nckap1*, NCK-associated protein; *Otx2*, orthodenticle homologue 2; *Pou5f1*, POU-domain class 5 transcription factor 1; TGFB, transforming growth factor- β .

a Pre-gastrulation: cell movement in the visceral endoderm



b Early gastrulation: genetic and molecular activity influencing the movement of the nascent mesoderm and the endoderm

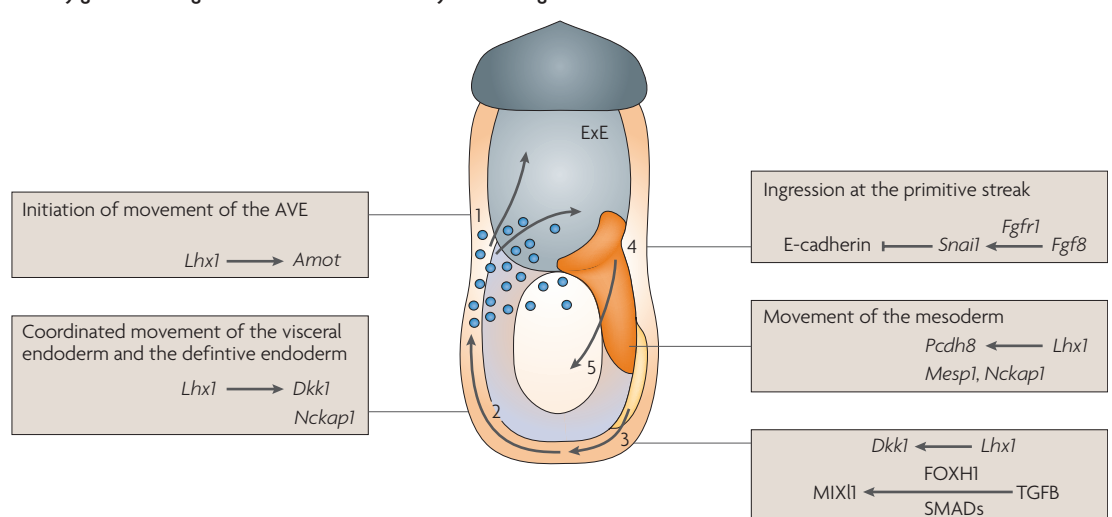


Figure 3 | Pattern of cell movement during embryogenesis and the molecular activities that influence cell or tissue movement. **a** | Patterns of cell movement in the visceral endoderm of the embryo before gastrulation. Distal visceral endoderm (DVE) cells move to the prospective anterior side of the embryo (indicated by a dashed arrow) to form the anterior visceral endoderm (AVE, indicated by blue dots) (step 1), lateral movement of the AVE after the leading population reaches the border (indicated by a dashed line) of the epiblast and the extraembryonic ectoderm (ExE) (step 2), and distribution of the visceral-endoderm cell clones (indicated by yellow dots) that are derived from the inner cell mass of the blastocyst in an oblique domain in the visceral endoderm (step 3) — this pattern can be seen at E5.5 (before the movement of the DVE) and at E6.5 at the onset of gastrulation⁶⁴. The scheme shows the enhancing factors, the driving forces and the hierarchy of transcription activities initiated by *Nodal* and *Gdf3* (growth differentiation factor 3) on putative *Nodal*/*Foxh1* (forkhead box 1) downstream genes for regulating DVE movement (see also TABLE 1). **b** | Movement of the mesoderm and endoderm at early gastrulation. Movement of the endoderm is accomplished by the coordination of movement of the AVE (again, indicated by blue dots) to the extraembryonic yolk sac (step 1), the anterior–proximal movement of the more distally located visceral endoderm (step 2) and anterior displacement of the definitive endoderm (shown in yellow) (step 3). Movement of the mesoderm (shown in orange) commences with the ingression of the progenitor cells through the primitive streak (step 4), followed by the anterior spreading of the tissue sheet (step 5). The scheme of genetic and molecular activities that affect the movement of specific cell populations in embryos was made on the basis of analysis of downstream gene activity (in the direction of pointers shown in the boxes) and putative functional interactions (positive interactions are indicated by an arrow; inhibitory interactions are indicated by a bar pointer). *Amot*, angiominin; *APC*, adenomatous polyposis coli protein; *Dkk1*, dickkopf 1; *FGF*, fibroblast growth factor; *Lhx1*, LIM-domain homeobox 1; *Mesp1*, mesoderm posterior 1; *Mixl1*, MIX1 homeobox-like 1; *Nckap1*, NCK-associated protein 1; *Otx2*, orthodenticle homologue 2; *Pcdh8*, protocadherin 8; *TGFB*, transforming growth factor-β.

Morula

A stage of pre-implantation development in which the embryo is a ball containing 8–32 cells inside the zona pellucida. The outer cells will become the trophoblast and the inner cells will become the inner cell mass, from which the epiblast and subsequently the fetal tissues will form.

Fate mapping

An experimental approach by which the developmental fates of cells are traced from the tissue of origin to their destination and their contribution to specific types of tissues in the embryo is assessed.

Primitive endoderm

An extraembryonic cell layer that forms on the surface of the inner cell mass and faces the blastocoelic cavity. It gives rise to the visceral and parietal endoderm.

Blastocoele

A fluid-filled cavity within the blastocyst.

Epithelium

Cells that line the surface of a structure, characterized by tight cell junctions and polarized morphology.

Regionalization

The acquisition of localized morphological or molecular characteristics by cell populations within a tissue or an organ.

ectoplacental cone in contact with the epiblast, the DVE is still formed, indicating that the ectoplacental cone might supply the necessary signals for DVE formation⁴². Knockdown of *Bmp4* (bone morphogenetic protein 4) expression in the whole embryo showed that BMP4 might be one of the signals that is required for the normal formation of the DVE⁴³ (TABLE 1). However, whether this is a direct effect of BMP4 on the visceral endoderm or mediated by signals from the epiblast is unknown.

The involvement of WNT signalling from the epiblast in DVE formation is hinted at by the phenotype of embryos that lack β -catenin (encoded by *Ctnnb1*) or APC (adenomatous polyposis coli protein, which modulates canonical WNT signalling) (TABLE 1). Loss of *Ctnnb1* function in the epiblast and the constitutive β -catenin activity that is associated with the *Apc^{min}* mutation⁴⁴ both result in impaired DVE formation owing to defective proximal–distal regionalization of the visceral endoderm^{38,45}. By contrast, the loss of *Wnt3* does not affect the formation of the AVE⁴⁶, indicating that DVE formation could be mediated by other WNT factors or driven by WNT-independent β -catenin activity.

Nodal signalling, which originates in the epiblast (FIG. 2a, step 6) and is transduced by SMAD2 in the extraembryonic tissues, is required for the formation of the DVE^{47,48}. The identification of the nodal co-receptor teratocarcinoma-derived growth factor (encoded by *Tdgf1*, also known as *Cripto*) as a candidate β -catenin downstream target points to a potential link between nodal and WNT signalling in DVE formation⁴⁹, but loss of *Tdgf1* seems to affect the ability of the DVE to move, rather than its initial specification⁵⁰. It has been suggested that nodal signalling might work by supporting epiblast growth until there is sufficient separation between the distal EmVE and the ExE, in order to attenuate some as yet undefined inhibitory signals from the ExE for the specification of the DVE¹². Nodal alone can induce the expression of DVE marker genes even when the ExE is removed, indicating that nodal signalling might function in both overcoming inhibitory signals from the ExE and inducing DVE formation independently of the ExE¹².

Although the DVE is formed in embryos that are homozygous for a deletion of *Foxh1* (forkhead box H1, an inducer of nodal-dependent target genes, also known as *Fast2*), expression of *Lefty1* is absent in the DVE, consistent with a role for nodal in activating DVE genes⁵¹. Intriguingly, *Lefty1* expression is also lost in the DVE of mutant embryos that have epiblast-specific abrogation of *Foxh1* activity. As autoinduction of *Nodal* requires FOXH1, loss of *Lefty1* expression in these mice might be due to delayed or reduced autoinduction of *Nodal*, resulting in a reduced signal concentration.

Epiblast patterning

The formation of the primitive streak is an important milestone in embryogenesis (FIG. 1e). It is the morphological indication of the onset of gastrulation, which results in the formation of the primary germ layers and the establishment of the basic body plan of the embryo. Formation and correct placement of the primitive streak requires the

prior establishment of anterior–posterior polarity in the epiblast and the overlying visceral endoderm.

It has been proposed that the formation of the primitive streak is accompanied by the posterior migration of cells that express primitive-streak marker genes³³. However, a recent study shows that patterns of gene expression do not necessarily reflect the migratory behaviour of the prospective primitive-streak cells. For example, expression of *T* (brachyury) is shifted to the posterior epiblast before posterior movement of epiblast cells is detected³⁶.

Reciprocal interactions between the epiblast and the ExE maintain the signals that are required for formation of the primitive streak. Signals from the ExE, including BMP4, induce expression of primitive-streak markers^{40,42,43,52–54} (FIG. 2b; TABLE 1). In a reciprocal interaction, nodal signalling from the epiblast maintains *Bmp4* expression in the ExE, which probably occurs independently of nodal processing (FIG. 2b, step 1)^{47,55}.

A correct balance between the activity of nodal and its antagonists in the AVE is necessary for the anterior–posterior patterning of the epiblast and correct placement of the primitive streak^{47,56,57}. Abrogation of early nodal signalling in the epiblast results in a failure of primitive-streak gene expression to shift to the posterior epiblast^{51,58}. Analysis of embryos that express a cleavage-resistant nodal pro-protein shows that *Fgf8* expression in the primitive streak might be induced by the uncleaved nodal protein⁵⁵. The nodal antagonists CER1 and LEFTY1, which are produced in the DVE and later the AVE, attenuate the influence of nodal signalling (FIG. 2b, step 4). In embryos that lack both antagonists, the anterior side of the primitive streak becomes expanded, or ectopic primitive streaks form. *Cer1/Lefty1*-deficient embryos that lack one functional *Nodal* allele exhibit a less severe phenotype, confirming that correctly regulated nodal signalling is necessary for epiblast patterning⁵⁶.

Primitive-streak formation requires specification of the posterior epiblast by WNT3 signalling through β -catenin and LRP5 (low-density lipoprotein receptor-related protein 5) and LRP6 (REFS 16,45,59,60). Expression of *Wnt3* is activated by the unprocessed form of nodal, probably through BMP4 signalling in the ExE^{47,55} (FIG. 2b, steps 6,7). In turn, WNT3 activates a feedback loop that maintains nodal expression in the epiblast⁵⁵. However, *Wnt3*-dependent activity does not account for all the nodal downstream activity that affects primitive-streak formation. This indicates that, although WNT signalling works downstream of nodal signalling in primitive-streak formation, other WNT-independent nodal functions are required for this process.

What drives cell movement in the embryo?

Movement of cells in the embryo is accomplished by a combination of active cell-autonomous migration and the bulk movement of the tissue⁶¹. Responses to environmental cues for inducing and directing cell movement require interaction between cellular mechanisms of locomotion and signal transduction systems. Morphogenetic cell and tissue movements have a vital role in shaping

the embryo, but the mechanisms that direct cell movements have been elusive. Recent studies have revealed potential roles of differential cell proliferation, active guidance and homotypic interactions in directing cell movement in the visceral endoderm and germ-layer progenitors.

Active versus driven movement of the DVE cells. An important step in establishing the primary body axes is the generation of anterior–posterior asymmetry. Signalling through the BMP, nodal and WNT pathways is initially located at the proximal end of the epiblast, whereas antagonists emanate from the DVE (FIG. 2a,b). Translocation of the DVE to the anterior side^{62,63} (FIG. 3a) transforms this proximal–distal polarity into the prospective anterior–posterior axis. However, specification of anterior–posterior polarity might take place earlier in view of the asymmetry in gene expression in the PrE of the implanting blastocyst and the asymmetrical distribution of the clonal descendants of ICM cells in the visceral endoderm⁶⁴ (see FIG. 3a, step 3). Although this has yet to be proven, it is possible that DVE movement might be a morphogenetic response to a pre-existing anterior–posterior polarity.

Active migration of the DVE cells (FIG. 3a, step 1) has been tracked using a *Hhex*–GFP reporter⁶⁵. The migratory ability of cells is influenced by dynamic changes in the cytoskeleton that are mediated by WASF1 (Wiskott–Aldrich syndrome protein family member 1, also known as WAVE). WASF1 can constitutively activate the actin-nucleating complex (ARP2/3), enabling the cell to perceive the environmental cues for migration⁶⁶. WASF1 forms a complex with NCK-associated protein 1 (NCKAP1 or NAP1), enhancing its stability and membrane localization⁶⁶. Loss of *Nckap1* results in loss of polarized morphology and no movement of the DVE⁶⁶.

DVE migration can be directed by nodal and WNT signalling activity (FIG. 3a). The DVE fails to move in many mutant embryos, including those that lack the function of certain components of these two signalling pathways (TABLE 1). A potential role of WNT activity in DVE movement is shown by manipulating the activity of *Dkk1* in *Otx2*^{−/−} (orthodenticle homologue 2) mutant embryos⁶⁷. The DVE is formed in the *Otx2*-null mutant but fails to move^{34,67}. The *Otx2*-deficient DVE can be mobilized by expressing DKK1 (Supplementary information S1 (table)) as a knock-in allele at the *Otx2* locus, or by reducing the level of WNT/β-catenin activity. Several WNT genes are expressed in the pre-gastrula embryo⁶⁸, but which of these is involved with cell movement is not known. WNT3 is unlikely to have a role, as visceral-endoderm movement is not affected in the *Wnt3*^{−/−} embryo¹⁶. Experimentally, movement of the DVE can be directed by localized DKK1 and WNT3A activity, which generate attractive and repulsive action, respectively. The DVE can also be driven towards and away from the site of ectopic expression of *Lefty1* and *Nodal*, respectively. It is proposed that nodal activity elevates cell proliferation on the prospective posterior side of the embryo, whereas the inhibition of nodal by CER1 reduces cell proliferation in the anterior side. This might result in localized build-up of cells, generating the

propulsive force that directs DVE movement³⁷. However, regionalized accretion of cells alone cannot account for the rapid movement of DVE cells, which reach the border of the epiblast and the ExE within 4–5 hours⁶⁵ (FIG. 3a, step 1), or the subsequent lateral movement of the AVE cells (FIG. 3a, step 2). It is more likely that the differential rate of cell proliferation that is induced by nodal is responsible for sustaining the driven movement of the visceral endoderm, combined with the WNT-regulated active movement of the visceral-endoderm cells.

Cell movement through the primitive streak. The formation of the germ layers requires movement of the progenitor cells from the epiblast through the primitive streak, to either emerge as a new mesoderm layer or be incorporated as definitive endoderm into the pre-existing visceral-endoderm layer (BOX 1). The crucial role of FGF signalling in regulating cell movement is highlighted by the phenotypic effect of mutations of genes that encode FGF8 and FGFR1 (fibroblast growth factor receptor 1, see FIG. 3b, step 4).

In *Fgfr1*-mutant embryos, epiblast cells fail to undergo the epithelium-to-mesenchyme transition, which is required for ingression through the primitive streak⁶⁹. This defect is most likely attributable to the failure to upregulate *Snail* (snail homologue 1) and to down-regulate *Cdh1* (cadherin 1, also known as E-cadherin) in these mutants. *Fgfr1*^{−/−} embryos also have reduced expression of the T-box genes *Tbx6* and *T*, which affects the movement of the mesoderm. Expression of *T* is indirectly regulated by FGFR1 through an interaction with WNT signalling. In *Fgfr1*^{−/−} cells with high E-cadherin activity, alterations in the physical association between E-cadherin and β-catenin reduce nuclear localization of β-catenin, which results in downregulation of *T*, a direct target of WNT/β-catenin signalling^{69,70}. A null mutation of *Fgf8* causes the accumulation of mesenchyme in the vicinity of the primitive streak⁷¹, which is consistent with a role for FGF8 in promoting dispersion of ingressed cells from the primitive streak in the chick⁷².

Recent findings point to a crucial role for homotypic interactions between cell-surface complexes in guiding cell movement during gastrulation. A screen for WNT downstream genes by expression profiling of β-catenin mutant embryos uncovered the gene that encodes IFITM1 (interferon inducible transmembrane protein 1, also known as *Fragilis* 2 or *Mil2*), which is expressed in the pre-migratory primordial germ cells (PGC) in the epiblast and the embryonic mesoderm^{73,74}. IFITM1 is a component of the cell-surface protein complex for homotypic cell–cell interactions. PGCs that do not express *Ifitm1* are repelled by *Ifitm1*-expressing primitive streak cells, resulting in translocation of the PGC to the endoderm, which does not express *Ifitm1*⁷⁴. The molecular mechanism that mediates the repulsive activity between cells with different IFITM1 activity is unknown. The activity of another IFITM gene, *Ifitm3* (also known as *Fragilis* or *Mil1*), which might be subject to induction by BMP⁷⁵, seems to enable the cells to perceive the homing cue for localization to the correct sites for the PGCs⁷⁴. It is not known if the IFITM1-mediated repulsive action also

Homotypic interactions
Interactions between cells with similar cell-surface characteristics or molecular properties.

Epithelial-to-mesenchyme transition

A process during which cells lose their epithelial characteristics, downregulate E-cadherin, gain a less regular appearance and become migratory.

Primordial germ cells

Diploid germline precursors that migrate through the gut endoderm layer and enter the gonad, where they differentiate into germ cells.

navigates the movement of other *Ifitm1*-negative cells, which could include the progenitors of the definitive endoderm, during germ-layer formation.

Movement of the mesoderm and endoderm at gastrulation. *Lhx1* (LIM-domain homeobox 1, also known as *Lim1*) has a important role in the morphogenetic movement of the mesoderm and the endoderm. During gastrulation, AVE cells are dispersed to the extraembryonic region (FIG. 3b, step 1), but in the *Amot*^{-/-} (angiomotin) mutant these cells remain stationary⁷⁶. As *Amot* expression is regulated by *Lhx1*, movement of the endoderm² (FIG. 3b, steps 2,3) in the *Lhx1*^{-/-} embryo might be blocked by the AVE cells that do not move. Loss of *Lhx1* also affects the movement of the mesoderm (see FIG. 3b, step 5), accompanied by the downregulation of **PAPC** (paraxial protocadherin, encoded by *Pcdh8*)⁷⁷. In *Xenopus laevis*, downregulation of PAPC leads to upregulation of C-cadherin and affects cell sorting during gastrulation⁷⁸. PAPC also interacts with Frizzled 7 receptor to activate the RhoA–JNK pathway, and might therefore function through the non-canonical WNT pathway to control cell movement^{79,80}. However, in the mouse, whether *Lhx1* function in the mesoderm movement is mediated by PAPC⁵¹ or has any link to WNT signalling is not known.

Movement of the mesoderm and endoderm is not obligately coupled. Embryos that lack *Mesp1* (mesoderm posterior 1) function exhibit retarded movement of the mesoderm but not the endoderm⁸¹. In the *Mixl1*^{-/-} (MIX1-homeobox-like 1) embryo, in which fewer definitive endoderm cells are recruited to the endodermal layer, a reduction in propulsion might underpin the lack of endoderm migration that occurs despite the movement of underlying mesoderm⁸² (FIG. 3b).

The impact of endoderm movement on patterning. The regulation of cell movement before and during gastrulation is crucial, not only for shaping the embryo, but also for positioning the tissue to achieve normal inductive interactions for morphogenesis. A common feature of three types of mutant embryo that we have discussed is the duplication of the posterior body axis^{66,83,84}. An expansion of the expression domain of *Gsc* (goosecoid, which marks the anterior segment of the primitive streak), *T* and *Wnt3* (which both mark the whole primitive streak) is found in the *Lhx1*, *Mixl1* and *Nkap1* mutant embryo, respectively. Loss of *Nkap1* affects the movement of both the visceral and definitive endoderm, whereas loss of *Lhx1* and *Mixl1* affects the movement of the definitive endoderm (FIG. 3b). Consistent with the role of the visceral endoderm in the specification and positioning of the primitive streak (see the previous section^{56,85}) and the role of the definitive endoderm in patterning the body axis⁸⁶, the abnormal movement of the endoderm might cause axis duplication by disrupting the normal spatial pattern and temporal course of tissue interaction.

Signals regulate lineage choices at gastrulation

The progenitors of different mesoderm and endoderm tissues are allocated during gastrulation (BOX 1; FIG. 4a).

The formation of different populations of mesoderm and endoderm cells seems to depend on the deployment of different levels of nodal and WNT signalling activity in the primitive streak (FIG. 4a,b; **Supplementary information S2** (table)).

Loss of nodal signalling leads to the absence of anterior primitive-streak (APS) derivatives, such as the axial and paraxial mesoderm and the definitive endoderm^{37,87–89}. Conversely, elevation of nodal signalling due to the loss of the transcription co-repressor or nodal antagonists (**Supplementary information S2** (table)) result in the overexpression of mesodermal markers^{56,57,90}. The generation of different mesoderm and endoderm derivatives might be a response to exposure to different intensities or durations of nodal signalling⁵⁵. This is highlighted by differing effects on the formation of APS-derivatives when nodal signalling is disrupted to varying degrees in embryos that have different combinations of *Smad2* and *Smad3* alleles, or that lack the *Nodal* proximal epiblast enhancer^{91,92}. Complete loss of nodal signalling prevents germ-layer formation. Progressively smaller reductions in nodal signal transduction in embryos result in varying degrees of loss of axial and paraxial mesoderm, with the mildest reduction causing loss of the prechordal plate only, but not the rest of the axial mesoderm^{91,92}. Taken together, these findings indicate that formation of the anterior derivatives of the primitive streak requires a high intensity of nodal signals, and that there is a graded response to nodal signalling in the allocation of different mesodermal derivatives. Whether a gradient of nodal signalling activity exists in the primitive streak (FIG. 4a) is not known.

The requirement of high levels of nodal for formation of definitive endoderm is supported by the observation that endodermal marker genes are activated in mouse ES cells when high concentrations of a TGFB-related factor, activin A, are present⁹³. A similar effect is observed with human ES cells⁹⁴, indicating a conserved role for SMAD2/3-mediated TGFB signalling in exercising the lineage choice for endoderm. Although specification of the definitive endoderm requires high nodal activity, its formation after the initial step of specification depends on activation of *Mixl1* to downregulate nodal⁸⁴. The *Mixl1* promoter contains SMAD- and FOXH1-binding sites, which might mediate the response to TGFB signalling⁹⁵. *Nodal* expression is elevated in the *Mixl1*^{-/-} mutant, indicating that MIXL1 might have a negative-feedback action (FIG. 4b).

Other TGFB-related factors, along with WNT and FGF (**Supplementary information S2** (table))^{96–98}, are involved in the specification of germ-layer tissues (**Supplementary information S2** (table)). *Bmp4*^{-/-} embryos are deficient in posterior derivatives of the primitive streak, such as the extraembryonic and haematopoietic mesoderm⁹². Conditional ablation of *Bmpr1a* (bone morphogenetic protein receptor, type 1A) in the epiblast results in excessive production of paraxial mesoderm⁹⁹. This defect can be rescued by lowering the FGF signalling activity, indicating that BMP and FGF activities counteract each other in the allocation of paraxial mesoderm. Like the action of TGFB/nodal antagonists, amnionless

Paraxial mesoderm

The precursor of the somites, which give rise to the spine, muscle and dermis of the skin.

Prechordal plate

A mesoderm-derived structure that lies underneath the forebrain and anterior to the notochord.

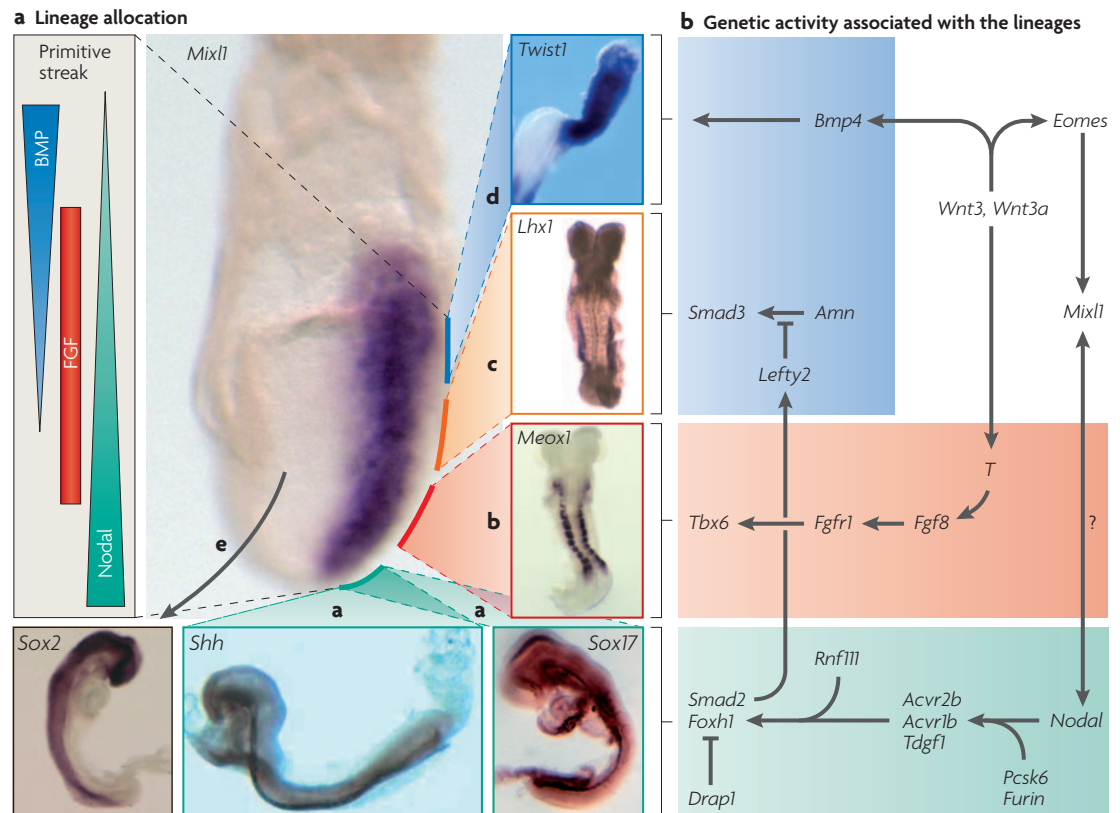


Figure 4 | Gene expression and regionalized allocation of tissue progenitors in the primitive streak during gastrulation. **a** | Allocation of epiblast cells in the anterior-most region of the primitive streak (the node) to the axial mesoderm (expressing *Shh* (sonic hedgehog)) and the definitive endoderm (expressing *Sox17* (SRY-box containing gene 17)) (box **a**), the anterior-third segment of the primitive streak to the paraxial mesoderm (expressing *Meox1* (mesenchyme homeobox 1)) (box **b**), the middle-third segment of the primitive streak to the lateral plate mesoderm (expressing *Lhx1* (LIM-domain homeobox 1)) (box **c**) and the posterior-third segment of the primitive streak of progenitor of the extraembryonic mesoderm (for example, allantois, expressing *Twist1*) (box **d**). Cells that remain in the epiblast are allocated to the ectoderm (for example, neuroectoderm, which expresses *Sox2*) (box **e**). The putative patterns of graded nodal and BMP (bone morphogenetic protein) activity and regionalized FGF (fibroblast growth factor) signalling activity in the primitive streak are shown schematically, with the width of the bar indicating the strength of activity. **b** | Genetic activity (colour-matched with the signalling activity) mediates the specification or allocation of the tissue progenitors in the four regions of the primitive streak (the brackets). Genes in the relational network are those that, when mutated, are associated with defects in the formation of specific tissues. The connectors indicating an inter-relationship are based on known functional hierarchy and experimental demonstration of epistatic (up- or downstream) relation, but not transcriptional regulation. The question mark denotes an as yet undefined interaction that is related to transcriptional regulation. *Acvr2b*, activin receptor IIB; *Acvr1b*, activin receptor IB; *Amn*, amnionless; *Drap1*, DR1-associated protein 1; *Eomes*, eomesodermin homologue; *Lefty1*, left-right determination factor 1; *Lhx1*, LIM-domain homeobox; *Mixl1*, MIX1 homeobox-like 1; *Pcsk6*, pro-protein convertase subtilisin/kexin type 6; *Rnf111*, ring finger 111 (also known as arkadia); *T*, brachyury; *Tdgf1*, teratocarcinoma-derived growth factor; *Tbx6*, T-box 6.

(TABLE 1; [Supplementary information S1,S2](#) (tables)), which might negatively regulate BMP signalling, is required for the formation of the paraxial and lateral-plate mesoderm, which are derived from the middle region of the primitive streak^{56,100}. In addition, formation of the paraxial mesoderm, which is derived from both the anterior and middle primitive streak, requires WNT signalling. In *Wnt3a*-null mutant mice, ectopic neuroectoderm is formed in place of somites¹⁰¹. The formation of middle-primitive-streak derivatives is therefore subject to the combined action of TGFB, FGF and WNT signalling¹⁰², whereas that of the posterior primitive streak depends on BMP activity (FIG. 4a,b).

Conclusions

A comparison of development prior to gastrulation in different groups of vertebrates shows that the degree of conservation is, to an extent, dependent on the mode of reproduction and development that is used in that species. Although the mouse develops from a blastocyst of similar configuration to many mammals, the architecture of the post-implantation embryo (FIG. 1) is unique to the rodents¹⁰³. The interactions between the extraembryonic and embryonic tissues take on a much greater importance in mammalian development than in many species that have less extensive development of extraembryonic structures.

To facilitate comparisons between these developmental stages in the mouse and other systems, it is useful to conceptually flatten the cup-shaped 'egg cylinder' into a discoid configuration, with the ectoplacental cone removed. In this model, the epiblast is depicted at the centre of the disc surrounded by a ring of ExE, with the visceral endoderm lying underneath². This configuration shows that the tissues on the periphery of the epiblast (equivalent to the proximal epiblast of the cup-shaped embryo) will be most strongly influenced by the ExE, because of their close proximity. In this way, the regionalization of the visceral endoderm and the interactions between the DVE or AVE and overlying regions of the epiblast disc can be visualized more clearly^{2,104}. This view of the mouse embryo places the visceral endoderm in an equivalent location to the hypoblast of the chick or rabbit. The hypoblast of the rabbit, like the mouse visceral endoderm, develops molecular asymmetries along the anterior–posterior axis before the epiblast exhibits anterior–posterior polarity¹⁰⁵. In the chick, primitive-streak formation requires the interaction of TGFB-related molecules, such as nodal and VG1, and WNT signals^{106–108}. Signalling activity is modulated by WNT and nodal antagonists from the hypoblast, similar to that in the mouse DVE and AVE, for correct positioning of the primitive streak^{105,106}. The hypoblast of chick embryos therefore functions like the DVE and AVE of the mouse as a source of inhibitory and inductive activity¹⁰⁹.

Genetic investigations of mouse mutants^{11,12,55,56,98} and analysis of the transcriptome^{110,111} and downstream gene activities using mutants^{49,73,112}, undertaken in conjunction

with experimental embryology^{37,40,41,63,65,67}, have revealed complex morphogenetic interactions between extraembryonic and embryonic tissues. These interactions are accomplished through the coordination of interdependent signalling pathways that are often reciprocal and autoregulatory in nature. The challenge now is to gain a holistic understanding of the multitude of processes that interact to regulate embryogenesis, by analysing gene functions in whole embryos as well as cell- and tissue-based experimental models. Recent studies using mouse and human ES-cell differentiation as a model have highlighted the conservation of signalling function in maintaining pluripotency and regulating cell-lineage choice^{13,94}. However, one must not forget that the mouse gastrula has adopted a configuration that is not typical of the mammals¹⁰³, and therefore the genetic and molecular paradigms for mouse embryogenesis must be road-tested in other embryonic models. The conservation of developmental controls must therefore be examined in other species with different embryonic morphology. For example, the role of inductive interaction between the juxtaposed ExE and epiblast in tissue patterning and that of the AVE in determining the axis, as shown in the mouse, might need to be verified in the guinea pig embryo, in which the ExE is not in direct contact with the epiblast¹¹³, and in the armadillo conceptus, which produces obligate quadruplets in which the anterior poles of the body axes converge centripetally¹⁰³, respectively. Outcomes of the study of comparative embryology will tell us whether there is a common developmental paradigm for mammalian embryogenesis.

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Competing interests statement

The authors declare no competing financial interests.

DATABASES

The following terms in this article are linked online to:

Entrez Gene: <http://www.ncbi.nlm.nih.gov/entrez/query.fcgi?db=gene>
Bmp4 | *Cer1* | *Dab2* | *Fgf4* | *Gata6* | *Gsc* | *Hnf4a* | *Lefty1* | *Lhx1* | *Mixl1* | *POU5F1* | *T* | *TdGF1* | *Wnt3*
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 β -catenin | *DKK1* | *GRB2* | *GSK3* | *FGFR1* | *nanog* | *NCKAP1* | *Nodal* | *PAPC* | *SALL4* | *SMAD2* | *SMAD3* | *SOX2*

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